

NEW RICHMOND, WI, USA

WMO: 722179

Lat: 45.150N

Lon: 92.533W

Elev: 996

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-9.0	-21.6	1.7	-12.8	-17.3	2.2	-8.6	24.1	15.2	21.0	15.7	4.3	300	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	88.0	71.7	84.3	69.9	81.6	68.3	75.0	83.4	73.1	80.9	71.3	78.2	11.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	125.1	79.1	70.5	116.6	77.0	69.0	110.4	75.4	39.2	83.7	37.4	81.7	35.7	78.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.7	19.1	17.2	DB	-21.8	93.0	4.9	3.2	-25.4	95.3	-28.2	97.2	-31.0	99.0	-34.6	101.4	
			WB	-22.2	78.4	4.8	2.3	-25.6	80.1	-28.5	81.5	-31.2	82.8	-34.7	84.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	14.5	16.7	30.9	44.0	56.4	65.8	70.3	67.6	61.2	47.4	33.9	19.8
	DBStd	21.94	13.12	12.04	12.60	9.84	8.90	5.94	5.50	5.61	7.98	8.91	10.21	11.65
	HDD50	4498	1101	933	605	228	36	0	0	0	10	158	490	937
	HDD65	8059	1566	1353	1059	630	293	60	15	36	163	549	933	1402
	CDD50	2372	0	0	12	49	234	473	628	545	345	79	7	0
	CDD65	457	0	0	0	1	27	82	178	115	49	5	0	0
	CDH74	3928	0	0	3	14	286	673	1594	938	384	36	0	0
	CDH80	976	0	0	0	0	75	147	451	220	81	2	0	0
	WSAvg	7.4	7.8	7.8	7.8	9.0	8.2	7.0	5.9	5.5	6.7	7.5	8.0	7.7
	PrecAvg	33.1	0.9	0.8	1.7	2.9	3.9	4.8	4.4	4.2	3.6	3.0	1.8	1.0
Precipitation	PrecMax	42.8	2.2	1.7	3.3	7.5	7.7	8.2	8.6	7.8	7.7	6.5	5.0	2.2
	PrecMin	25.1	0.2	0.1	0.5	0.7	1.0	1.9	2.2	0.9	0.9	1.0	0.0	0.3
	PrecStd	4.6	0.6	0.4	0.7	1.5	1.8	1.6	1.7	1.9	1.7	1.8	1.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.7	49.8	70.5	76.3	88.0	89.0	92.6	89.8	87.3	78.9	65.9	46.2
		MCWB	36.5	45.8	58.3	58.2	66.6	71.4	74.3	71.9	69.4	64.1	54.2	42.7
	2%	DB	36.9	40.7	61.0	70.0	81.0	84.2	89.3	85.7	81.9	72.8	58.6	41.0
		MCWB	34.4	36.2	52.1	54.6	64.0	68.6	74.2	70.8	68.8	60.8	50.4	37.7
	5%	DB	34.4	36.6	54.1	65.8	76.9	80.8	84.8	82.0	78.7	67.8	53.6	36.8
		MCWB	32.9	33.4	47.0	51.0	61.2	66.3	71.4	68.4	66.2	58.5	46.8	34.8
	10%	DB	32.2	33.1	46.7	61.0	72.1	78.0	81.7	79.3	74.8	61.9	48.5	34.3
		MCWB	31.1	30.7	41.3	48.6	58.8	64.6	69.5	67.0	65.0	53.2	43.2	33.1
	0.4%	WB	37.1	44.0	60.5	60.1	71.9	75.2	79.4	76.3	74.7	68.5	56.9	45.5
		MCDB	40.0	49.5	68.3	69.8	84.0	85.3	85.3	84.2	81.0	73.1	63.0	45.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.8	36.8	54.1	56.7	67.3	71.8	76.1	73.8	71.6	63.4	51.7	37.8
		MCDB	36.8	39.8	59.3	67.4	75.9	79.9	84.7	81.5	78.2	71.0	58.0	40.1
	5%	WB	33.0	33.6	47.0	53.3	64.3	69.2	73.7	72.0	69.3	58.4	46.8	35.3
		MCDB	34.2	36.3	54.2	62.1	71.9	76.6	82.0	77.9	74.8	65.4	51.7	36.8
	10%	WB	31.1	31.2	42.0	50.3	61.6	67.2	71.8	70.0	66.9	54.8	43.5	33.2
		MCDB	32.1	32.8	46.9	59.6	69.1	74.7	79.6	75.7	73.2	60.6	48.8	34.0
Mean Daily Temperature Range	5% DB	MDBR	15.7	17.7	17.5	19.2	20.3	19.0	20.1	19.9	20.1	17.7	14.8	13.6
		MCDBR	14.7	17.9	23.3	27.8	26.2	22.4	22.6	22.5	22.9	24.6	21.7	14.8
		MCWBR	12.6	14.4	14.3	15.4	12.2	10.6	10.6	10.1	11.7	14.0	14.5	12.6
	5% WB	MCDBR	13.9	16.7	21.5	24.2	20.7	18.7	19.7	18.7	18.3	20.0	19.8	12.8
		MCWBR	12.2	13.7	14.0	14.9	11.6	10.8	11.0	10.6	11.1	13.6	14.0	11.1
Clear-Sky Solar Irradiance	taub		0.258	0.281	0.318	0.367	0.400	0.392	0.384	0.395	0.366	0.319	0.286	0.264
	taud		2.399	2.323	2.345	2.309	2.264	2.323	2.369	2.347	2.401	2.519	2.508	2.422
	Ebn at Noon		277	288	290	283	276	278	279	271	271	272	264	261
	Edh at Noon		23	30	34	39	42	40	37	37	32	25	21	20
All-Sky Solar Radiation	RadAvg		543	888	1224	1463	1688	1875	1983	1656	1288	812	516	379
	RadStd		60	89	107	145	136	142	67	117	88	108	47	44

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	-3.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page